

Barath R

Embedded Engineer

Chennai, India | +91-9442169681 | barath5727@gmail.com

github.com/barath0508 | linkedin.com/in/barath0508 | barath0508-portfolio.vercel.app

Professional Summary

Third-year Electronics and Communication Engineering student and Embedded Engineer with hands-on proficiency across hardware design, bare-metal firmware, and software engineering. Experienced in building from silicon to software – from register-level RISC-V assembly and custom PCB/sensor bring-up to full-stack IoT telemetry pipelines, cloud integration, and algorithmic problem solving. Seeking an Embedded Engineer role to architect high-performance, connected hardware and software systems.

Education

B.E. Electronics and Communication Engineering	2024 – 2028 (Expected)
Rajalakshmi Institute of Technology, Chennai	CGPA: 8.65/10

Currently in 3rd year

Higher Secondary Certificate (HSC) – Class XII	2023 – 2024
Montfort Matriculation Higher Secondary School, Chennai	Percentage: 95%

Tamil Nadu State Board

Secondary School Leaving Certificate (SSLC) – Class X	2021 – 2022
Montfort Matriculation Higher Secondary School, Chennai	Percentage: 94%

Tamil Nadu State Board

Technical Skills

- **Hardware & Embedded Systems:** ESP32, ESP32-C6, ESP32-CAM, Raspberry Pi 5, Arduino UNO R3, Altera DE2 FPGA, Bare-Metal MMIO, CST Studio, Keysight ADS
- **PCB Design & EDA Tools:** PCB/PCD Design, KiCad, Proteus, EasyEDA, Schematic Capture, Multi-Layer PCB Layout, Routing, DRC/ERC, Gerber & BOM Generation, Component Footprints, SMD/THT Soldering, Hardware Bring-up
- **Hardware Protocols & Interfacing:** I2C, SPI, UART, MQTT, LoRaWAN, GPIO, PWM, ADC, MMIO, WebSockets, HTTP/REST
- **Programming Languages:** C, C++, Python, JavaScript (ES6+), TypeScript, Java, SQL, RISC-V Assembly
- **Software & Web Development:** React.js, Next.js, Node.js, Express.js, FastAPI, Flask, HTML5, CSS3, Tailwind CSS, REST APIs
- **Cloud & Databases:** Firebase, HiveMQ, ThingSpeak, AWS IoT Core, Google Cloud Platform, MongoDB, PostgreSQL, MySQL, SQLite
- **Engineering Tools & DevOps:** Git, GitHub Actions, Docker, Linux/Bash, ESP-IDF (v6), VS Code, PlatformIO, MATLAB, Scilab, ADB
- **Core CS & Problem Solving:** Data Structures & Algorithms (LeetCode streak), OOP, Concurrency, RTOS Architecture

Featured Projects

Embedded Systems & Hardware Engineering

Bare-Metal RISC-V Trap Handler & Context Switcher	ESP32-C6 / Bare-Metal
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Tech Stack: RISC-V Assembly, C, Direct MMIO, Hardware Timers, Interrupt Controller

- Designed and implemented a bare-metal trap handler and timer-based preemptive context switcher in assembly with no RTOS or vendor libraries
- Configured memory-mapped I/O (MMIO) registers for hardware interrupt handling, stack frame preservation, and register-level task scheduling

ESP-Drone – Autonomous Flight Controller & Firmware	ESP32-C6 / Quadcopter Firmware
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Tech Stack: C/C++, ESP-IDF, FreeRTOS, IMU Sensor Fusion, PID Control, ESP-NOW / Wi-Fi, MCPWM

- Engineered real-time flight control firmware for an autonomous micro-quadcopter powered by the ESP32-C6 RISC-V microcontroller
- Implemented 6-DOF IMU sensor fusion with complementary filtering, quaternion attitude estimation, and automated failsafe routines
- Developed high-frequency 500 Hz dual-loop PID stabilization algorithms with high-resolution motor PWM and low-latency ESP-NOW telemetry

Digital Twin for Industrial Monitoring	5G-Integrated IoT System
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Tech Stack: ESP32 / Edge Node, Multi-Sensor Fusion, 5G Telemetry, Python, Real-Time Web Dashboard

- Architected an industrial digital twin connecting multi-sensor edge nodes with 5G connectivity for real-time equipment telemetry and predictive maintenance
- Developed cloud data ingestion pipelines and responsive dashboard visualization, bridging sensor-level data acquisition to software analytics

Dual-Axis Solar Tracker

Embedded Control System

Tech Stack: Custom PCB Prototyping, Microcontroller, Servo Motors, LDR Sensor Array, Closed-Loop Control, C/C++

- Constructed a dual-axis solar tracker optimizing photovoltaic panel orientation using an analog LDR sensor bridge and closed-loop feedback
- Designed custom sensor interfacing and motor driver circuitry with voltage regulation; programmed precision tracking algorithms in C/C++

Conformal Helmet-Integrated Antenna Array

RF & Defense Hardware

Tech Stack: Microstrip Patch Array, Dual-Band UHF/L-Band, Flexible PCB Substrate, CST Microwave Studio, ADS, RF Shielding

- Engineered a flexible microstrip patch antenna array conformal to curved helmet geometry, replacing bulky whip antennas for tactical gear
- Modeled dual-band UHF/L-band radiation patterns in CST/ADS and integrated an RF shielding layer with ruggedized coax interfacing

Software & Full-Stack Web Engineering

iCreatePDF

100% Client-Side Privacy PDF Suite

Tech Stack: Next.js 16, TypeScript, WebAssembly (WASM), Web Workers, Tailwind CSS

- Architected a 100% client-side privacy-first document suite featuring 37+ PDF tools running entirely in-browser with zero server uploads
- Implemented high-throughput in-memory execution using WebAssembly and Web Workers for sub-second compression, merging, and booklet imposition
- Deployed production web application at icreatepdf.online featuring responsive UI, dark mode, and offline PWA capability

QRLoop

Dynamic QR & Scan Analytics Platform

Tech Stack: React.js, Node.js, Express.js, Supabase (PostgreSQL), JWT, HTML5 Canvas

- Engineered a full-stack dynamic QR platform enabling post-print destination redirection without reprinting physical media
- Built real-time scan analytics capturing device telemetry, referrers, and geolocation with rate-limited JWT authentication
- Developed custom HTML5 canvas rendering engine supporting brand color palettes, custom logos, and dot-matrix visual styling

FlashESP

Web Serial Firmware Flasher & In-Browser IDE

Tech Stack: Next.js, Web Serial API, esptool.js, Monaco Editor, Supabase, Tailwind CSS

- Built an in-browser ESP32 firmware flasher and editor eliminating desktop driver and toolchain dependencies via Web Serial API and esptool.js
- Integrated Monaco Editor with C/C++ syntax highlighting, real-time serial baud rate monitoring, and interactive terminal telemetry

Energize 2026 Hackathon Portal

Interactive 3D Web Application

Tech Stack: React.js, TypeScript, Three.js, GSAP, Web Audio API, Vercel

- Engineered official symposium and hackathon web portals featuring interactive 3D digital twins and dynamic models rendered with Three.js
- Implemented GSAP timeline choreography, magnetic cursor interactions, ambient Web Audio API state machine, and registration flows

Technical Experience

- Conceptualized, engineered, and deployed embedded and full-stack solutions end-to-end, spanning custom PCB fabrication, bare-metal firmware, high-throughput cloud telemetry, and responsive web applications
- Architected open-source client-side and full-stack software products (iCreatePDF, QRLoop, FlashESP), adopting WebAssembly, TypeScript, and modern API architectures
- Executive Technical Organizer at IETE Students' Forum (ISF), coordinating technical hackathons and workshops across software and hardware tracks